

INTERNAL STUDY CODE: VV_CC-PT/30D3T1(D)1_723_22_001

Study Report - Version nº 4

**ASSESSMENT IN HUMAN OF THE CUTANEOUS COMPATIBILITY OF A COSMETICAL
PRODUCT AFTER A SINGLE UNDER PATCH APPLICATION UNDER DERMATOLOGICAL
CONTROL**



Reception date: August 18th 2022	Experimental phase end date: September 02nd 2022
Experimental phase start date: August 29th 2022	Report date: September 20th 2022

SPONSOR

S&J INTERNATIONAL ENTERPRISES CO., LTD.

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Information provided by the sponsor:

Tested product

Product name: KJELDGAARD BARRIER DEFENSE
Product reference: FORMULATION NO.SBW006-05
Batch: -

Zurko Research S.L. is not responsible for the data provided by the sponsor, included in this table.

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1. SYNOPSIS

Internal study code	VV_CC-PT/30D3T1(D)1_723_22_001
Panelists	Number of panelists at the beginning: 33 Age range: 18-70 years All skin types (sensitive and non-sensitive skin) Number of panelists at the end: 32
Test area	Upper back
Application	Duration: 5 days Frequency: only once
Test period	August 29th 2022 - September 02nd 2022
Test parameter	Cutaneous evaluation of erythema and edema
Study design	Day 1 (0 hours) – Sample preparation and application Day 5 (96 hours) – Clinical and dermatological evaluation
Evaluation	Cutaneous Mean Irritation Index (M.I.I.)
Results	Under adopted experimental conditions, the product KJELDGAARD BARRIER DEFENSE , reference: FORMULATION NO.SBW006-05 is Non irritating (NI) . In conclusion, it has Good cutaneous compatibility .

2. STUDY OBJECTIVE AND PRINCIPLE

The study was carried out following general conditions in Zurko Research, established for the execution of study project on humans (*Structure and content of clinical study reports from ICH Harmonised Tripartite Guideline Topic E3; Guideline for good clinical practice E6(R2) of June 14th 2017; Regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014*).

This study has as an objective verifying the cutaneous compatibility of a cosmetic product after a single application on the skin under exaggerated experimental conditions.

The product was applied only once over the skin of the back under a patch.

The compatibility of the product with the skin was verified, after 15-30 minutes of removing the patches and by means of visual exam of the experimental area, by the responsible technical expert, as well as by a dermatologist in charge for the study.

The negative control excluded false positives.

3. PANELISTS

3.1. Ethical aspects

Each panelist participating in the study has been informed before about the type and the procedures of the study, signing an Informed Consent Form before the beginning of the study.

3.2. Specific inclusion and exclusion criteria

The specific inclusion criteria, defined in the protocol, were as follow:

- Age: 18-70 years old
- Photo-type (Fitzpatrick): I to IV
- All skin types (sensitive and non-sensitive skin)

The specific exclusion criteria, defined in the protocol, were as follow:

- Cutaneous marks on the experimental area that could interfere with the evaluation of the skin reactions (pigmentation disruptions, scars, excessive hair areas, excessive freckles and moles, solar skin burns, tattoos...).
- Injuries, pathologies or infection in the experimental area.
- Eczematous reaction which has not fully disappeared, scar or pigmentation complications from previous studies in the experimental area.
- Intention to bath in the bath, swimming pool or the sea, or having sauna or Turkish baths during the study.
- Intense sun or UV ray exposure during the study or during the previous month to the study.

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- Carrying out a treatment containing acid vitamin A or its by-products, during the 3 months previous to the study.
- Carrying out a treatment containing topical corticoids, on the experimental area during the 8 days previous to the beginning of the study.
- Treatment with any medicine for psoriasis, vitiligo, within one month before the study.
- Vaccination prediction during the study period or having had the last vaccine within the 3 weeks previous to the study.
- Being pregnant or breastfeeding.
- Allergies to metals.
- Reactivity to medical tape.
- Participation during the previous 30 days in any study under exaggerated conditions (under a patch).

Information about the panelists is included in Annex I.

4. MATERIALS AND EQUIPMENT

- Finn Chamber Aqua® occlusive patch and/or Curatest® semioclusive patch
- Pasteur pipettes 1ml, syringes 2 ml
- Sterile container
- Distilled water, sanitary alcohol 70º
- Cotton, tissues
- Wood depressors, tweezers, scissors
- Gloves
- Precision Balance Model: PS 750. R2. Radwag

5. METHODOLOGY

5.1. Application characteristics

Product type: leave on gel.

Product preparation: sample is not diluted.

Applied quantity: 20 µl or mg of product preparation over occlusive patch (Finn Chamber Aqua® occlusive patch).

5.2. Experimental procedure

	D1	D5
Signature of the Informed Consent Form	X	
Specific inclusion and exclusion criteria	X	
Product application	X	
Clinical assessment (individual results in Annex II)		X

5.3. Interpretation of Results

The analysis of the results was carried out according to the results obtained under the experimental conditions, for this purpose the Mean Irritation Index (M.I.I.) was calculated. The M.I.I. value was used to classify the product according to Table 1.

$$M.I.I. = \frac{\sum(\bar{x} \text{ erythema and edema grade})}{\text{Number of panelists}}$$

M.I.I.	Product Classification
M.I.I.=0,000	Non-Irritating (NI)/Very Good Cutaneous Compatibility
M.I.I.<0,200	Non-Irritating (NI)/Good Cutaneous Compatibility
0,200≤M.I.I.<0,500	Slightly Irritating (SI)/Intermediate Cutaneous Compatibility
0,500≤M.I.I.<1,000	Moderately Irritating (NI)/Bad Cutaneous Compatibility
M.I.I.≥1,000	Irritating (I)/Very Bad Cutaneous Compatibility

Table 1. Product classification according its global M.I.I.

6. RESULTS

Next table shows the M.I.I. at 15-30 minutes after patch removal.

M.I.I.	Results	Number of reactive panelists	Reactive panelists %
0,016	Non irritating (NI) / Good cutaneous compatibility	1	3%

7. CONCLUSIONS

The results obtained in the laboratory and included in this report correspond to the sample analyzed in the laboratory.

Under adopted experimental conditions, the product **KJELDGAARD BARRIER DEFENSE**, reference: **FORMULATION NO.SBW006-05** is **Non irritating (NI)**. In conclusion, it has **Good cutaneous compatibility**.

8. MODIFICATIONS

Version	Description	Date
01	Report implementation	September 13th 2022
02	Product reference correction	September 19th 2022
03	Product reference correction	September 19th 2022
04	Product name correction	September 20th 2022

New report versions replace the previous one.

9. DEVIATIONS

There were no deviations during the study.

10. SAMPLES AND DOCUMENTS TO BE STORED

The documentation related to the study will be stored in the facilities of Zurko Research.

Documents relating to the study are stored for 5 years. After this time the client will be informed to know how to proceed. In case of no response, they will be destroyed.

The tested product will be stored in the Zurko Research sample library for 1 year. After this time, it will be eliminated through the usual waste management procedure for this type of product.

11. BIBLIOGRAPHICAL REFERENCES

1. SCCS (Scientific Committee on Consumer Safety), SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 11th revision, 30-31 March 2021, SCCS/1628/21.
2. Levin, C.Y., H.I. Maibach. "Animal, Human, and In Vitro Test Methods for Predicting Skin Irritation". Dermatotoxicology, Chpt. 44; 7th Ed., F.N. Marzulli; H.I. Maibach; Taylor and Frances.
3. Holdiness, M.R., 1989, "A Review of Contact Dermatitis Associated with Transdermal Therapeutic Systems". Contact Dermatitis, 20(1);3-9.
4. North American Contact Dermatitis Group Patch-Test Results, 2001-2002 Study Period. Dermatitis: December 2004. Marks, James G. Jr; Belsito, Donald V.; DeLeo, Vincent A.; Fowler, Joseph F. Jr; Fransway, Anthony F.; Maibach, Howard I.; Mathias, Toby C.G.; Nethercott, James R.; Rietschel, Robert L.; Rosenthal, Lawrence E.; Sherertz, Elizabeth F.; Storrs, Frances J.; Taylor, James S.
5. Nueva Clasificación de tipos piel y sus implicaciones en Dermatología Cosmética. Revisión Dermatología Venezolana. Vol. 43, Nº 4, 2005. Leslie Baumann, Sadegh Amini, Eduardo Weiss.

12. SIGNATURES

The undersigned declare that the study was carried out following general conditions in Zurko Research, established for the execution of study project on humans (*Structure and content of clinical study reports from ICH Harmonised Tripartite Guideline Topic E3; Guideline for good clinical practice E6(R2) of June 14th 2017; Regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014*).

The results here presented reflect accurately and completely the raw data of the study.

Main researcher: Jesús González Cuartero, declare that this study has been carried out under my responsibility.

Dermatology Team: Zurko's dermatology team, led by the dermatologist Javier Pedraz (medical license number: 283706434), and Natalia Zawierta, as adjunct dermatologist, declare that this study has been reviewed under their responsibility. In representation,

Annex I: Information relating to panelists

Panelist	Age	Sex	Skin type	Phototype	Withdrawal/Exclusion
V01	28	F	S	III	Exclusion: positive control
V02	54	F	R	IV	No
V03	66	F	R	IV	No
V04	39	F	S	IV	No
V05	29	F	R	III	No
V06	23	F	S	IV	No
V07	45	M	S	II	No
V08	62	F	R	III	No
V09	44	F	R	IV	No
V10	52	F	R	IV	No
V11	32	F	S	II	No
V12	22	F	S	IV	No
V13	44	F	R	IV	No
V14	52	F	S	III	No
V15	46	F	S	IV	No
V16	28	F	S	IV	No
V17	69	F	S	IV	No
V18	19	M	S	IV	No
V19	48	F	S	III	No
V20	22	M	R	III	No
V21	22	F	S	IV	No
V22	56	F	R	II	No
V23	29	F	S	III	No

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V24	41	F	S	III	No
V25	40	F	R	III	No
V26	53	F	S	III	No
V27	49	M	R	III	No
V28	67	M	S	III	No
V29	43	F	R	IV	No
V30	63	F	R	IV	No
V31	56	F	S	III	No
V32	28	F	R	IV	No
V33	61	F	R	III	No

F: female; M: male; R: non-sensitive; S: sensitive

Annex II: Individual results at 15-30 minutes after patch removal

Panelist	Erythema result T96	Erythema control T96	Edema result T96	Edema control T96	Other cutaneous reaction (D; DT; T; R)
V02	0	0	0	0	-
V03	0	0	0	0	-
V04	0	0	0	0	-
V05	0	0	0	0	-
V06	0	0	0	0	-
V07	0	0	0	0	-
V08	0	0	0	0	-
V09	0	0	0	0	-
V10	0	0	0	0	-
V11	0	0	0	0	-
V12	0	0	0	0	-
V13	0	0	0	0	-
V14	0	0	0	0	-
V15	0	0	0	0	-
V16	0	0	0	0	-
V17	0	0	0	0	-
V18	1	0	0	0	-
V19	0	0	0	0	-
V20	0	0	0	0	-
V21	0	0	0	0	-
V22	0	0	0	0	-

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V23	0	0	0	0	-
V24	0	0	0	0	-
V25	0	0	0	0	-
V26	0	0	0	0	-
V27	0	0	0	0	-
V28	0	0	0	0	-
V29	0	0	0	0	-
V30	0	0	0	0	-
V31	0	0	0	0	-
V32	0	0	0	0	-
V33	0	0	0	0	-
M.I.I. diario / daily	0,031		0,000		

0: non-visible erythema; 0,5: very slight erythema; 1: slight erythema, 2: moderate erythema, 3: intense erythema
D: dryness, DT: detergency, T: thickness, R: reflectivity